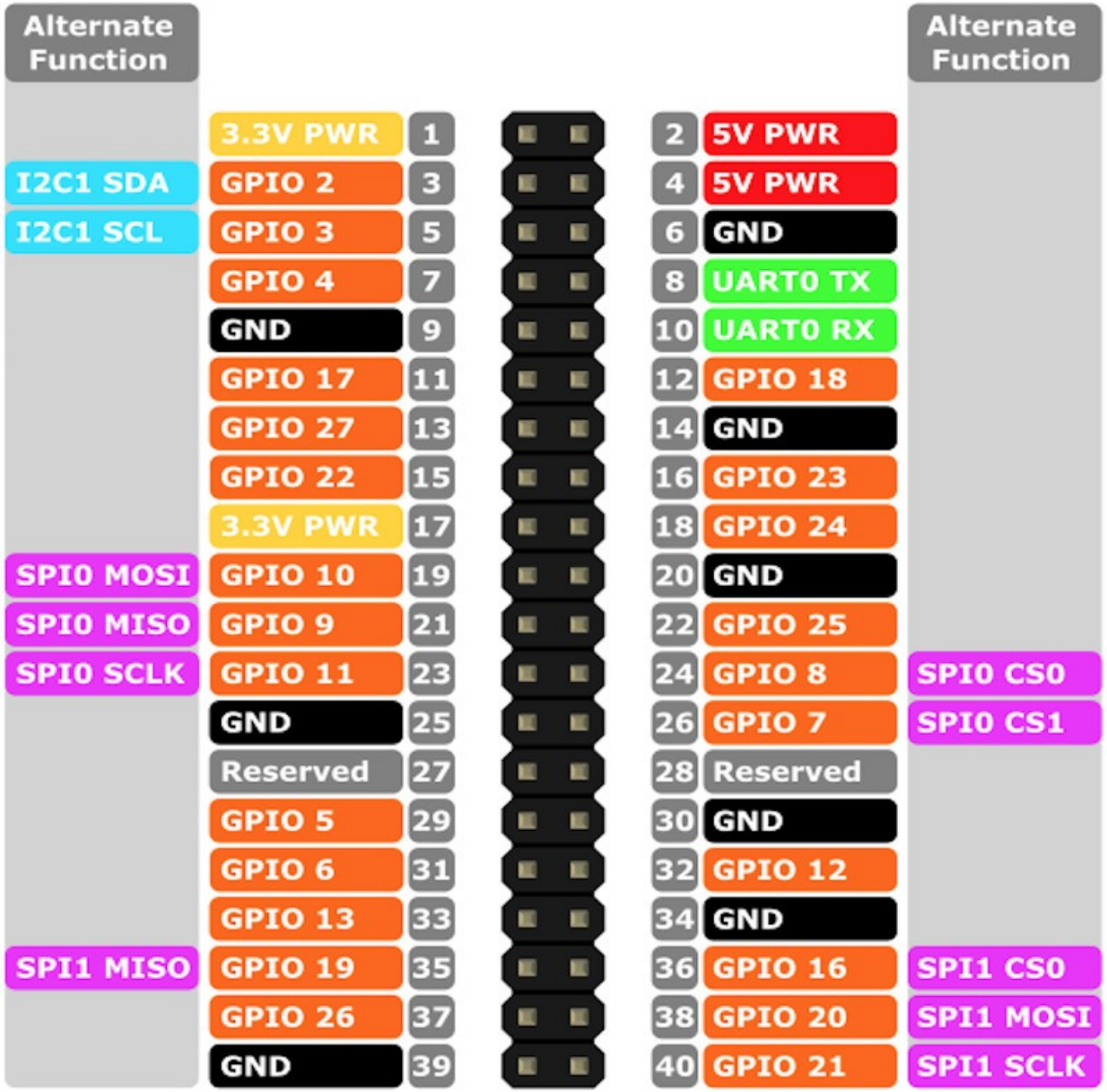
Raspberry Pi 4 GPIO Pins



- Raspberry Pi is developed by Raspberry Pi Foundation in the United Kingdom. The Raspberry Pi is a series of powerful, small **single-board computers.**
- Various versions of Raspberry Pi have been out till date.
- All versions consist of a Broadcom system on a chip (SoC) with an integrated ARM-compatible CPU and on-chip graphics processing unit (GPU).
- The original device had a single-core Processor speed of device ranges from 700 MHz to 1.2 GHz and a memory range from 256 MB to 1 GB RAM.
- To store the operating system and program memory SD cards are used. Raspbian OS which is a Linux operating system is recommended OS by Raspberry Pi Foundation. Some other third party OS like RISC OS Pi. Diet Pi, Kali, Linux can also be used.
- It also provides a set of general purpose input/output pins allowing to control electronic components for physical computing and explore the Internet of Things (IOT).

Diagram of Raspberry-Pi

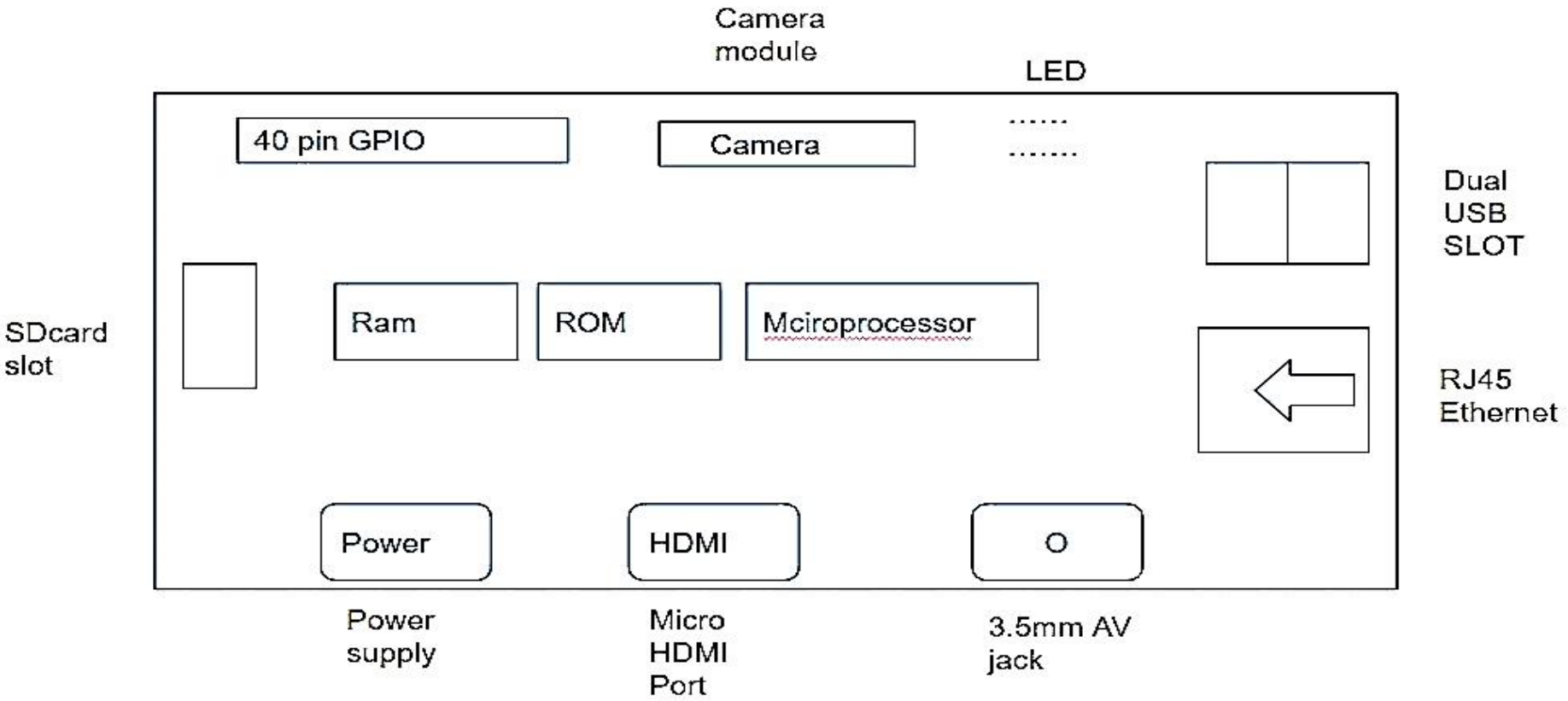


Fig : Raspberry pi

- **Raspberry Pi model –**
- There have been many generations of raspberry Pi from Pi 1 to Pi4. There is generally a model A and model B. Model A is a less expensive variant and it trends to have reduce RAM and dual cores such as USB and Ethernet.
- **List of Raspberry pi models and releases year:**
- pi 1 model B – 2012
- pi 1 model A – 2013
- pi 1 model B+ -2014
- pi 1 model A+ – 2014
- Pi 2 Model B – 2015
- Pi 3 Model B- 2016
- Pi 3 Model B+ -2018
- Pi 3 Model A+ -2019
- Pi 4 Model A – 2019
- Pi Model B – 2020
- Pi 400 – 2021
- Pi 5 -2023

- **Specs of the Computer: –** The computer has a quad-core ARM processor that doesn't support the same instruction as an X86 desktop CPU. It has 1GB of RAM, One HDMI port, four USB ports, one Ethernet connection, Micro SD slot for storage, one combined 3.5mm audio/video port, and a Bluetooth connection. It has got a series of input and output pins that are used for making projects like – home security cameras, Encrypted Door lock, etc.
- **Versatility of Raspberry Pi: –** it can be used for watching videos on YouTube, watching movies, and programming in languages like Python, Scratch, and many more. it has a series of I/O pins that give this board the ability to interact with its environment and hence can be utilized to build robotics and automation projects.
- **Examples of projects: –** It can be turned into a weather station by connecting some instruments to it for check the temperature, wind speed, humidity etc. home surveillance system due to its small size; by adding some cameras to it the security network will be ready.

Blink LED code in rpi

```
from gpiozero import LED
import time
a = LED(17)
while True:
    a.off()
    time.sleep(1)
    a.on()
    time.sleep(1)
```

Button to switch on LED rpi code

```
from gpiozero import Button, LED
import time
led = LED(17)
b = Button(2)
b.wait_for_press()
led.on()
time.sleep(2)
led.off()
```

Switch to toggle LED in RPi

```
from gpiozero import LED, Button
import time
led = LED(17)
button = Button(2)
while True:
    button.wait_for_press()
    led.toggle()
    time.sleep(1)
```

IR sensor display

```
from gpiozero import InputDevice
import time
sensor = InputDevice(17)
while True:
    if sensor.is_active:
        print("nothing detected")
        time.sleep(0.1)
    else:
        print("obstacle detected")
        time.sleep(0.1)
```

IR with LED

```
from gpiozero import InputDevice
import time
sensor = InputDevice(17)
while True:
    if sensor.is_active:
        print("nothing detected")
        time.sleep(0.1)
    else:
        print("obstacle detected")
        time.sleep(0.1)
```

Rpi code for Ultrasonic Senssor

```
from gpiozero import DistanceSensor

from time import sleep

sensor = DistanceSensor(trigger=17, echo=27)


try:

    while True:

        # The sensor.distance property returns the value in meters

        distance_m = sensor.distance

        distance_cm = round(distance_m * 100, 2)

        print(f"Distance: {distance_cm} cm")

        sleep(1)
```